

Cost reduction imperative to attain grid parity

Solar PV business is here to stay. The question is how soon it is going to become competitive to conventional energy, or how soon it is going to attain grid parity. There is going to be explosive growth in demand for solar PV with the attainment of grid parity.

Factors affecting grid parity are:

1. Cost of materials
2. Conversion efficiency
3. Cost of conventional energy

In India, all three factors are unfavourable compared to western (developed) world, but we have:

1. Huge market waiting to absorb economically reasonable energy alternatives
2. Massive pool of well-educated workforce
3. Nascent, favourable policy environment

The most reliable (with more than 25 years of field-performance data) solar PV technology available today is crystalline silicon. Therefore crystalline silicon technology is most suitable for adoption in India. However, thin-film technologies have the potential for phenomenal cost reduction and should therefore be kept active on the agenda for intense R&D efforts coupled with limited manufacturing presence.

The cost of crystalline silicon PV is heavily dependent on the cost of polysilicon. There were abnormal fluctuations in the price of polysilicon until recently due to excessive demand with limited supply. With the capacity augmentation efforts of existing players and entry of several new entrants, prices are now settling down to reasonable, stable levels. India is yet to see any significant, indigenous poly-manufacturing base. This has to be kept on the long-term government agenda for the growth of solar PV in India.

Next to polysilicon are the raw materials and consumables. Chief among them are wafering consumables, metalisation pastes, semiconductor grade chemicals, speciality gases and modulating materials. Non-availability of local sources for proven brands of these items is another handicap for cost-reduction efforts of the PV industry.

Availability of latest machinery from suppliers on turnkey basis is not an issue these days, but keeping pace with the rest of the world in the chosen technology is a difficult task without strong in-house R&D or collaboration arrangements with national or international research institutes. Since the cost of electricity is low in India (compared to the developed countries), achievement of grid parity will take longer for India (grid parity is expected by 2020).

—Contributed by a senior Bharat Electronics Limited (BEL) official

solar cells and cover all stages of the value chain of solar PV manufacturing.

While the raw material for solar cells (wafers) is available in the international market, it is subject to volatility in price, which can lead to isolated incidents of pressure on availability. In the long term, having the raw material (polysilicon) production and wafers could insulate us from some of these risks.

All the equipment for solar cell and module manufacturing are imported, as there is no manufacturer for solar manufacturing equipment in the country.

"As far as skills are concerned, there is a shortage of skills for the solar energy sector, keeping in mind that the sector is only taking off now. In the immediate future, there is going to

be a shortage of qualified personnel at different nodes of the value chain from manufacturing to marketing to installation and commissioning and O&M as well as R&D," says Subramanya.

Production cost

The production cost of solar cells at present is in the range of 1.5 to 2 dollars per peak watt (\$1.5-\$2/Wp). It can be brought down through economies of scale resulting from larger volumes as well as through technological improvements leading to better yields and higher efficiency of conversion of sunlight into electrical current. This is the focus of technological research and innovation worldwide.

Costs can be brought down with development of new, lower-cost and less energy-intensive techniques for

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